

IME 775 – Assignment 2 Solutions (Chapter 3)

Classifiers and Vector Calculus

Part A: Image Classification and Decision Boundaries

Problem 1

Rasterization converts a 2-D image (matrix) into a 1-D vector by reading elements left-to-right, top-to-bottom.

$$\vec{x} = [10, 20, 30, 40, 50, 60, 70, 80, 90]^T$$

Dimensionality = $3 \times 3 = 9$.

Problem 2

Sample answer:

In a classification problem every input can be represented as a vector, equivalently a point in a high-dimensional feature space. Points belonging to the same class tend to form clusters. A **decision boundary** is a hypersurface that separates these clusters. To classify a new input we check which side of the boundary the point falls on.

- **Linear boundary (hyperplane):** sufficient when clusters are well separated (e.g., car vs. giraffe in the textbook example).
 - **Nonlinear boundary (curved surface):** needed when clusters overlap or interleave (e.g., horse vs. zebra requiring a hypersphere).
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Problem 3

Sample answer:

Model architecture selection — Choose a parameterized function family $q(\vec{x}; \vec{w}, b)$ (e.g., a linear model for simple tasks, a multilayer neural network for complex tasks). At this stage the parameters $\vec{w}, b$ are still unknown; only the functional form is fixed.

Model training — Estimate $\vec{w}, b$ so that the model output matches the known ground truth on the training data as closely as possible, typically by iteratively minimizing a loss function via gradient descent.

The two stages are sequential: architecture selection defines the search space; training searches within that space.

Problem 4

Model: $q(\vec{x}) = \vec{w}^T \vec{x} + b = 2x_0 - 3x_1 + 1$.

4a) $q(\vec{x}_0) = 2(1) - 3(2) + 1 = 2 - 6 + 1 = -3$.

4b) $q = -3 \leq 0 \rightarrow \text{Class B}$.

4c) $q(\vec{x}_1) = 2(3) - 3(1) + 1 = 6 - 3 + 1 = 4 > 0 \rightarrow \text{Class A}$.

Part B: Loss Functions and Partial Derivatives

Problem 5

i	$\hat{y}_i$	$\bar{y}_i$	$e_i = \hat{y}_i - \bar{y}_i$	e_i^2
1	0.5	1.0	-0.5	0.25
2	-0.3	-0.2	-0.1	0.01

3 0.8 0.6 0.2 0.04

$$L = 0.25 + 0.01 + 0.04 = \boxed{0.30}$$

Problem 6

$$L(w_0, w_1) = 2w_0^2 + 3w_1^2$$

6a)

$$\frac{\partial L}{\partial w_0} = 4w_0, \quad \frac{\partial L}{\partial w_1} = 6w_1$$

6b)

$$\nabla L = \begin{bmatrix} 4w_0 \\ 6w_1 \end{bmatrix}$$

6c) At $(1, -2)$:

$$\nabla L = \begin{bmatrix} 4(1) \\ 6(-2) \end{bmatrix} = \begin{bmatrix} 4 \\ -12 \end{bmatrix}$$

Problem 7

7a) $\|\nabla L\| = \sqrt{4^2 + (-12)^2} = \sqrt{16 + 144} = \sqrt{160} \approx 12.6491.$

7b) The gradient points in the direction of **steepest increase** of L .

7c) To decrease L most rapidly, move in the **negative gradient** direction: $-\nabla L = [-4, 12]^T$.

Problem 8

$$L(w_0, w_1) = w_0^2 + 4w_1^2$$

8a) $\nabla L = [2w_0, 8w_1]^T$. At $(3, 1)$: $\nabla L = [6, 8]^T$.

8b) First-order approximation:

$$\Delta L \approx \nabla L^T \Delta \vec{w} = [6, 8] \begin{bmatrix} -0.1 \\ 0.2 \end{bmatrix} = 6(-0.1) + 8(0.2) = -0.6 + 1.6 = \boxed{1.0}$$

8c) Exact:

$$L(3, 1) = 9 + 4 = 13, \quad L(2.9, 1.2) = 8.41 + 5.76 = 14.17$$

$$\Delta L_{\text{exact}} = 14.17 - 13 = \boxed{1.17}$$

8d) The approximation (1.0) is close to the exact change (1.17); the error of 0.17 arises from the neglected higher-order terms. For smaller displacements the first-order approximation would be even more accurate.

Part C: Gradient Descent

Problem 9

Sample answer:

Input: Loss function $L(\vec{w})$, initial weights $\vec{w}^{(0)}$, learning rate η .

Repeat:

1. Compute the gradient $\nabla L(\vec{w}^{(k)})$.
2. Update weights: $\vec{w}^{(k+1)} = \vec{w}^{(k)} - \eta \nabla L(\vec{w}^{(k)})$.
3. Evaluate $L(\vec{w}^{(k+1)})$.

Terminate when L is sufficiently small or $\|\nabla L\| \approx 0$.

The learning rate η controls step size: too large risks overshooting the minimum; too small makes convergence slow.

Problem 10

$$L(w_0, w_1) = 2w_0^2 + 3w_1^2, \nabla L = [4w_0, 6w_1]^T, \eta = 0.1.$$

Step k	$\vec{w}^{(k)}$	$\nabla L(\vec{w}^{(k)})$	$\vec{w}^{(k+1)}$	$L(\vec{w}^{(k+1)})$
0	[1.000, -2.000]	[4.000, -12.000]	[0.600, -0.800]	2.640
1	[0.600, -0.800]	[2.400, -4.800]	[0.360, -0.320]	0.566
2	[0.360, -0.320]	[1.440, -1.920]	[0.216, -0.128]	0.142

Starting loss: $L(\vec{w}^{(0)}) = 2(1) + 3(4) = 14$.

Problem 11

11a) Yes $-14 \rightarrow 2.640 \rightarrow 0.566 \rightarrow 0.142$; the loss decreases monotonically.

11b) With $\eta = 1.5$, the update steps would be very large. For instance, $w_0^{(1)} = 1 - 1.5(4) = -5$, which overshoots the minimum at $w_0 = 0$ by a wide margin. The weights would oscillate wildly and the loss could *increase*, causing divergence.

Problem 12

At a minimum, the function surface is flat — there is no direction in which L decreases further. Mathematically, $\nabla L = \vec{0}$ at a minimum because all partial derivatives vanish (the function is at a stationary point). In gradient descent the update $\vec{w}^{(k+1)} = \vec{w}^{(k)} - \eta \nabla L$ becomes $\vec{w}^{(k+1)} = \vec{w}^{(k)}$ when $\nabla L = \vec{0}$, so the weights stop changing. Therefore, $\|\nabla L\| \approx 0$ (or L ceasing to decrease) serves as a practical stopping criterion.

Part D: First-Order Taylor Approximation

Problem 13

13a) First-order approximation:

$$L(w + \Delta w) \approx L(w) + \Delta w \cdot \frac{dL}{dw}$$

13b) $f(x) = x^3$, expand around $x = 2$ with $\Delta w = 0.1$:

$$f(2) = 8, \quad f'(x) = 3x^2, \quad f'(2) = 12$$

$$f(2.1) \approx 8 + 0.1 \times 12 = \boxed{9.2}$$

13c) Exact: $f(2.1) = 2.1^3 = 9.261$.

$$\text{Approximation error} = |9.261 - 9.2| = \boxed{0.061}.$$

The error is small because $\Delta w = 0.1$ is reasonably small.

Problem 14

$$L(w_0, w_1) = w_0 w_1 + w_0^2$$

$$\frac{\partial L}{\partial w_0} = w_1 + 2w_0, \quad \frac{\partial L}{\partial w_1} = w_0.$$

14a) At $(1, 3)$: $\nabla L = [3 + 2, 1]^T = [5, 1]^T$.

14b)

$$\Delta L \approx [5, 1] \begin{bmatrix} 0.05 \\ -0.1 \end{bmatrix} = 0.25 - 0.1 = \boxed{0.15}$$

14c) Exact:

$$L(1, 3) = 3 + 1 = 4$$

$$L(1.05, 2.9) = (1.05)(2.9) + (1.05)^2 = 3.045 + 1.1025 = 4.1475$$

$$\Delta L_{\text{exact}} = 4.1475 - 4 = 0.1475$$

The first-order approximation (0.15) is very close to the exact value (0.1475); the error of 0.0025 comes from neglected higher-order terms.

Part E: Level Contours and the Gradient

Problem 15

$$L(w_0, w_1) = w_0^2 + w_1^2$$

15a) The level contours $w_0^2 + w_1^2 = c$ are **circles** centered at the origin with radius $\sqrt{c}$.

15b) $\nabla L = [2w_0, 2w_1]^T$. At $(3, 4)$: $\nabla L = [6, 8]^T$.

15c) The level contour through $(3, 4)$ is $w_0^2 + w_1^2 = 25$ (a circle of radius 5). The tangent to this circle at $(3, 4)$ is in the direction $[-4, 3]^T$. Check perpendicularity:

$$\nabla L \cdot \vec{t} = [6, 8] \cdot [-4, 3] = -24 + 24 = 0 \quad \checkmark$$

The gradient is perpendicular to the level contour.

15d) Since the gradient is perpendicular to the level contour and points outward (toward higher L values), moving along the gradient direction crosses level contours as quickly as possible — this is the direction of steepest increase. Moving in the *opposite* direction (negative gradient) gives steepest decrease.

Problem 16

$$L(w_0, w_1) = w_0^2 + 4w_1^2$$

16a) The level contours $w_0^2 + 4w_1^2 = c$ are **ellipses** centered at the origin (wider along the w_0 axis).

16b) $\nabla L = [2w_0, 8w_1]^T$. At $(2, 1)$: $\nabla L = [4, 8]^T$.

16c) The level contour through $(2, 1)$ is $w_0^2 + 4w_1^2 = 8$. Implicit differentiation:

$$2w_0 + 8w_1 \frac{dw_1}{dw_0} = 0 \implies \frac{dw_1}{dw_0} = -\frac{w_0}{4w_1} = -\frac{2}{4} = -0.5$$

Tangent direction: $[2, -1]^T$ (or any scalar multiple).

$$\nabla L \cdot \vec{t} = [4, 8] \cdot [2, -1] = 8 - 8 = 0 \quad \checkmark$$

16d) One gradient descent step with $\eta = 0.1$:

$$\vec{w}^{(1)} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} - 0.1 \begin{bmatrix} 4 \\ 8 \end{bmatrix} = \begin{bmatrix} 1.6 \\ 0.2 \end{bmatrix}$$

New loss: $L(1.6, 0.2) = 2.56 + 0.16 = 2.72$ (down from 8).